Answer key

Practice Test 1 (Regional level) · 50 questions · 28 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

Multiple-choice answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	B	A	D	B	D	C	D	B	A	A	D	B	B

15	16	17	18	19	20	21	22	23
A	A	D	B	B	A	C	D	A

Section I: Multiple choice

1. A. André-Marie Ampère

1. Current is measured in amperes (A).
2. The ampere is named after Ampère.

Note: A1 History/Units Named After People
App: no app: fact or concept

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2. B. Alessandro Volta

1. Volta stacked zinc and copper (or silver) discs with brine-soaked cloth.
2. He announced the pile in 1800.

Note: A1 History/Volta
App: no app: fact or concept

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3. A. Writing an algebra of logic in which values are true or false

1. Boole wrote The Laws of Thought (1854).
2. His algebra of logic is the base of digital logic.

Note: A1 History/Boole
App: no app: fact or concept

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4. D. Cycles per second

1. 1 Hz = 1 cycle per second (frequency).
2. US mains is 60 Hz.

Note: A1 History/Hertz
App: no app: fact or concept

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5. B. Electrons moved from the hair to the balloon

1. Only electrons move in solids.
2. The balloon gained electrons, so it is negative.
3. The hair lost electrons, so it is positive.

Note: A2 Static Electricity and Coulomb/Charging Methods and Electroscopes
App: no app: fact or concept

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6. D. Negative

1. The positive rod pulls electrons to the near side.
2. Ground the far side, remove the ground, then remove the rod.
3. Induction gives the charge OPPOSITE to the rod: negative.

Note: A2 Static Electricity and Coulomb/Charging Methods and Electroscopes
App: no app: fact or concept

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7. C. 1/9

1. F is proportional to $1/r^2$.
2. $(1/3)^2 = 1/9$.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: no app: fact or concept

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8. D. To stop a static spark that could ignite fuel vapor

1. Flowing fuel builds up static charge.
2. The cable lets the charge flow to ground.
3. No charge build-up means no spark near the vapor.

Note: A2 Static Electricity and Coulomb/Static Hazards
App: no app: fact or concept

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9. B. 120 V AC at 60 Hz

1. US outlets give 120 V rms AC.
2. The frequency is 60 Hz.

Note: A3 DC and AC/Household Wiring
App: no app: fact or concept

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10. A. Neutral wire

1. Narrow slot = hot.
2. Wide slot = neutral.
3. The round hole is ground.

Note: A3 DC and AC/Household Wiring
App: no app: fact or concept

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11. A. One quarter as large

1. $R = \rho L/A$ and $A = \pi d^2/4$.
2. Twice the diameter gives 4 times the area.
3. So R is 1/4 as large.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: no app: fact or concept

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12. D. They go out

1. A broken filament is an open circuit.
2. A series circuit has only one path.
3. No current flows, so all bulbs go out.

Note: A5 DC Circuits/Open and Short Circuits
App: no app: fact or concept

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13. B. In series; zero resistance

1. The current must pass through the meter: series.
2. An ideal ammeter has zero resistance, so it does not change the current.

Note: A5 DC Circuits/Meters and Measurement
App: no app: fact or concept

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14. B. Blue, gray, brown, gold

1. 6 = blue, 8 = gray.
2. Multiplier 10 = brown.
3. ±5% = gold.

Note: A5 DC Circuits/Resistor Color Codes
App: `c680` → `4-band: blue gray brown gold` *(checked)*

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15. A. In series with the hot wire

1. A fuse must be in series to carry the load current.
2. It goes in the hot wire, so a blown fuse leaves the load unpowered (safe).

Note: A6 Switches Relays Fuses Breakers GFCI/Fuses
App: no app: fact or concept

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16. A. The hot and neutral currents differ by about 5 mA

1. A GFCI compares the hot and neutral currents.
2. A difference means current leaks to ground (maybe through a person).
3. It trips at about 4-6 mA.

Note: A6 Switches Relays Fuses Breakers GFCI/GFCI
App: no app: fact or concept

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17. D. Two SPDT (3-way) switches

1. Two SPDT ("3-way") switches are joined by two traveler wires.
2. Either switch can change the light.

Note: A6 Switches Relays Fuses Breakers GFCI/3-Way and 4-Way Switching
App: no app: fact or concept

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18. B. 0.7 V

1. A silicon PN junction turns on at about 0.6-0.7 V.

Note: A7 Diodes and LEDs/PN Junction and I-V Curve
App: no app: fact or concept

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19. B. Anode, connect it to +

1. Long leg = anode (+).
2. The flat edge on the rim marks the cathode (-).

Note: A7 Diodes and LEDs/LEDs
App: no app: fact or concept

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20. A. 0

1. AND gives 1.
2. NAND inverts it: 0.
3. NAND is 0 only when all inputs are 1.

Note: A8 Digital Logic/Logic Gates and Notation
App: no app: fact or concept

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21. C. NAND and NOR

1. NAND alone (or NOR alone) can build every other gate.

Note: A8 Digital Logic/Universal Gates (NAND and NOR)
App: no app: fact or concept

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22. D. A′ + B′

1. Break the bar and change the sign.
2. (AB)' = A' + B'.

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t-(ab)=-a+-b` → `EQUAL (all 4 rows match)` *(checked)*

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23. A. Remove the resistor from the powered circuit (power off)

1. An ohmmeter uses its own small battery.
2. Outside power or parallel parts give a wrong reading.
3. Turn power off and lift one end of the part.

Note: A9 Hands-On and Meters/Multimeter Use
App: no app: fact or concept

Section II: Short answer and calculations

24. 3.4 N, repel

1. $F = kq_1q_2/r^2$.
2. $F = 8.99e9 \times 3.0e\text{-}6 \times 5.0e\text{-}6 / 0.20^2 = 3.4$ N.
3. Same signs, so they repel.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `q a3u b5u r.2` → `f Force (+ repel) = 3.37 N` *(checked)*

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25. 3.0×10^{13} electrons

1. $n = Q / e$.
2. $n = 4.8e\text{-}6 / 1.602e\text{-}19 = 3.0 \times 10^{13}$.

Note: A2 Static Electricity and Coulomb/Charge Basics
App: `a q4.8u` → `e Electrons = 2.9959E13 electrons` *(checked)*

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26. 170 V

1. $V_{peak} = \sqrt{2} \times V_{rms}$.
2. $\sqrt{2} \times 120 = 170$ V.

Note: A3 DC and AC/AC Characteristics
App: `w r120` → `p Peak (amplitude) = 169.7 V` *(checked)*

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27. 8.0 h

1. t = capacity / current.
2. t = 2400 mAh / 300 mA = 8.0 h.

Note: A3 DC and AC/Batteries and Cells
App: `b c2400 i300m` → `t Time (hours) = 8 h` *(checked)*

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28. 60 W

1. P = VI.
2. P = 120 V × 0.50 A = 60 W.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `o v120 i.5` → `p Power = 60 W` *(checked)*

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29. 13.1 Ω

1. P = V^2/R, so R = V^2/P.
2. R = 120^2 / 1100 = 13.1 Ω.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `o p1100 v120` → `r Resistance = 13.09 ohm` *(checked)*

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30. $1.76 (13.5 kWh)

1. E = 0.900 kW × 0.5 h/day × 30 days = 13.5 kWh.
2. Cost = 13.5 × $0.13 = $1.76.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `e p900 h.5 d30 r.13` → `c Cost = $1.76 (1.755)` *(checked)*
App: `e p900 h.5 d30 r.13` → `e Energy = 13.5 kWh` *(checked)*

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31. 0.67 Ω

1. A = πd^2/4 = π(0.80e-3)^2/4 = 5.03e-7 m^2.
2. R = ρL/A = 1.68e-8 × 20.0 / 5.03e-7 = 670 mΩ.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: `r p1.68E-8 l20 d.8m` → `r Resistance = 0.66845 ohm` *(checked)*

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32. 10 Ω

1. 1/R = 1/60 + 1/30 + 1/20 = 6/60.
2. R = 10 Ω.

Note: A5 DC Circuits/Series and Parallel Resistance
App: `s60/30/20` → `Req = 10 ohm` *(checked)*

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33. 22 kΩ, ±5%

1. red = 2, red = 2: 22.
2. orange multiplier = 10^3: 22 000 ohm.
3. gold = ±5%.

Note: A5 DC Circuits/Resistor Color Codes
App: `c rrod` → `= 22000 ohm` *(checked)*

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34. 8.16 V

1. $V_{out} = V \times R_2/(R_1+R_2)$.
2. 12.0 × 10 / 14.7 = 8.16 V.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `d v12 a4.7k b10k` → `o Vout (across R2) = 8.163 V` *(checked)*

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35. 27 bulbs

1. Each bulb: I = 65/120 = 542 mA.
2. 15 / 0.5417 = 27.69.
3. Round DOWN: 27 bulbs.

Note: A6 Switches Relays Fuses Breakers GFCI/Fuses
App: `F f15 v120 p65` → `Max loads: 27 (load 28 blows the fuse)` *(checked)*

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36. 260 Ω

1. $V_R = 6.0 - 2.1 = 3.9$ V.
2. R = 3.9 V / 0.015 A = 260 Ω.

Note: A7 Diodes and LEDs/LEDs
App: `l s6 f2.1 i15m` → `r Resistor = 260 ohm` *(checked)*

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37. 101101

1. 45 = 32 + 8 + 4 + 1.
2. Bits 32 16 8 4 2 1: 1 0 1 1 0 1.

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x45` → `bin 101101 (6 bits)` *(checked)*

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38. 10010 (= 18 in decimal)

1. 1011 = 11 and 0111 = 7.
2. 11 + 7 = 18 = 10010.
3. Or add bit by bit with carries.

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x1011+111` → `sum bin 10010` *(checked)*

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39. F = A

1. Factor A: F = A(B + B').
2. B + B' = 1.
3. F = A.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `tab+a-b` → `F = A` *(checked)*

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40. XOR; F = A′B + AB′ (A ⊕ B)

1. F = 1 only when the inputs are different.
2. That is XOR: F = A'B + AB'.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t0110` → `This is an XOR gate` *(checked)*

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41. 1.5 kΩ

1. Same current in series: I = 2.4 V / 1000 ohm = 2.4 mA.
2. R_x = 3.6 V / 2.4 mA = 1.5 kΩ.
3. Short way: $R_x = R_k \times V_x/V_k$.

Note: A9 Hands-On and Meters/Task - Mystery Resistor
App: `m a1k v2.4 w3.6` → `x Mystery R (Rx) = 1.5 kohm` *(checked)*

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42. 2.2 h

1. Battery energy = 3.0 Ah × 3.7 V = 11.1 Wh.
2. Charger power = 5.0 V × 1.0 A = 5.0 W.
3. t = 11.1 / 5.0 = 2.2 h.

Note: A9 Hands-On and Meters/Task - USB Charger
App: `z c3000 v3.7 s5 a1` → `t Charge time = 2.22 h` *(checked)*

Section III: Circuit problems

Problem A: series-parallel circuit (Questions 43–47)

A 12.0 V battery drives R1 = 100 Ω in series with R2 = 200 Ω and R3 = 300 Ω, which are in parallel with each other (see the figure). Use ideal wires and an ideal battery.

43. 220 Ω

1. R2 || R3 = 200 × 300 / 500 = 120 ohm.
2. Add R1: 100 + 120 = 220 ohm.

Note: A5 DC Circuits/Series and Parallel Resistance
App: `s100+200/300` → `Req = 220 ohm` *(checked)*

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44. 54.5 mA

1. I = V / R_{eq} = 12.0 / 220 = 54.5 mA.

Note: A5 DC Circuits/Series and Parallel Resistance
App: `N V1-0 12 1-2 100 2-0 200 2-0 300` → `V1-0 12: 54.55 mA out of +, gives 654.5 mW` *(checked)*

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45. 6.55 V

1. V across the parallel pair = I × 120 ohm.
2. 0.05455 × 120 = 6.55 V.
3. Check: R1 gets 5.45 V; 5.45 + 6.55 = 12.0 V.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `N V1-0 12 1-2 100 2-0 200 2-0 300` → `V1 = 12 V V2 = 6.545 V` *(checked)*

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46. 21.8 mA

1. I3 = 6.545 V / 300 ohm = 21.8 mA.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `N V1-0 12 1-2 100 2-0 200 2-0 300` → `R2-0 300: 6.545 V 21.82 mA 2>0 142.8 mW` *(checked)*

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47. 655 mW

1. P = VI = 12.0 × 0.05455 = 655 mW.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `N V1–0 12 1–2 100 2–0 200 2–0 300` → `V1–0 12: 54.55 mA out of +, gives 654.5 mW` *(checked)*

Problem B: logic circuit (Questions 48–50)

The logic circuit below has inputs A, B and C and output F.

48. F = AB + C′

1. Top gate: AND gives AB.
2. Bottom gate: NOT gives C'.
3. OR joins them: F = AB + C'.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `tab+–c` → `F = AB + C'` *(checked)*

---

49. 1

1. AB = 1 · 0 = 0.
2. C' = 1.
3. F = 0 + 1 = 1.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: by hand (read row 100 of the t table if you want)

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50. 3 rows: 001, 011, 101

1. F = 0 needs AB = 0 AND C = 1.
2. C = 1 rows: 001, 011, 101, 111.
3. Drop 111 (AB = 1): 3 rows.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `tab+–c` → `M(1,3,5)` *(checked)*
